

Study details

Authors

Doris Hglinger, André Nadler, Per Haberkant, Joanna Kirkpatrick, Martina Schifferer, Frank Stein, Sebastian Hauke, Forbes D. Porter, Carsten Schultz

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Abstract

Lipid-mediated signaling events regulate many cellular processes. Investigations of the complex underlying mechanisms are difficult because several different methods need to be used under varying conditions. Here we introduce multifunctional lipid derivatives to study lipid metabolism, lipid-protein interactions, and intracellular lipid localization with a single tool per target lipid. The probes are equipped with two photoreactive groups to allow photoliberation (uncaging) and photo-cross-linking in a sequential manner, as well as a click-handle for subsequent functionalization. We demonstrate the versatility of the design for the signaling lipids sphingosine and diacylglycerol; uncaging of the probe for these two species triggered calcium signaling and intracellular protein translocation events, respectively. We performed proteomic screens to map the lipid-interacting proteome for both lipids. Finally, we visualized a sphingosine transport deficiency in patient-derived Niemann-Pick disease type C fibroblasts by fluorescence as well as correlative light and electron microscopy, pointing toward the diagnostic potential of such tools. We envision that this type of probe will become important for analyzing and ultimately understanding lipid signaling events in a comprehensive manner.

Lipid probes utilized

Trifunctional Sphingosine

Trifunctional Diacylglycerol

Trifunctional 8-3 Fatty Acid

Cell line analyzed

[HeLa](#)

Species: *Homo sapiens*

Cell type: Epithelial (immortalized cervical cancer line)

Uncaging & Crosslinking timeline

Lipid Probe	Uptake time	Uncaging time	Interaction time	Crosslinking time
Sph	5 min	2.5 min	0 min	2.5 min
DAG	15 min	2.5 min	0 min	2.5 min
8-3 FA	15 min	2.5 min	0 min	2.5 min

Mass spectrometry quantification method

Peptide spectral counting